Federated Learning Assessment for ICU Research (FLAIR): A Model-to-Data Paradigm for Privacy-Preserving Clinical AI Evaluation

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BACKGROUND: Clinical prediction models developed at a single institution often fail when deployed elsewhere. A widely adopted sepsis algorithm, for example, demonstrated poor discrimination at external sites [1]. Robust generalizability testing requires evaluation across diverse patient populations, clinical practices, and health systems, yet current benchmarks rely on one or two public datasets that represent a narrow slice of ICU care. Accessing private institutional data for broader validation typically requires patient-level data sharing, a practice constrained by regulatory, legal, and institutional barriers [2, 3]. The model-to-data paradigm inverts this relationship: researchers contribute only their trained model, which executes locally against private patient data, and only aggregate performance metrics are returned [4]. This approach was first demonstrated for clinical mortality prediction in a crowdsourced benchmarking challenge [5]. We extend this paradigm with a standardized benchmarking platform that handles cohort construction and evaluation at each site, lowering barriers to piloting innovative prediction methods in healthcare, establishing clinically relevant benchmarks, and minimizing the distance between model developers and the bedside.

METHODS:

Framework Design. The framework is distributed as an open-source Python package anchored in a common longitudinal ICU data format (CLIF) adopted by a consortium of U.S. academic medical centers [6]. The package handles cohort construction, outcome labeling, temporal train/test splitting, and dataset extraction, converting raw vital signs, laboratory values, respiratory support parameters, medications, and assessments into standardized dataset at each site. The researcher owns the feature engineering and the model that generates predictions, while built-in evaluation functions assess performance in accordance with TRIPOD-AI reporting guidelines [9]: discrimination (AUROC, AUPRC), calibration (Brier score, integrated calibration index), and clinical utility (decision curve analysis) [8]. Only aggregate outputs leave the institution; site principal investigators review all results before release. A publicly available ICU dataset in the common format provides an open-source entry point, enabling method development on public data in CLIF format with guaranteed portability to the consortium sites [7].

Demonstration Study. We applied this novel framework to predict in-hospital mortality across eight U.S. academic medical centers. The study included adult ICU patients admitted between 2018 and 2024 who survived at least 24 hours. At 24 hours after ICU admission, we extracted demographics, vital signs, laboratory values, and respiratory support data. The primary outcome was in-hospital mortality. We trained XGBoost and ElasticNet classifiers at the lead site, then distributed model objects to seven external sites for local evaluation. We compared three deployment strategies: (1) cross-site validation using the lead site model on 2024 data at each external site, (2) transfer learning with local fine-tuning on 2018–2023 data, and (3) independent site-specific training on 2018–2023 data. We assessed discrimination (AUROC), calibration (Brier score), and clinical utility (net benefit at a 20% decision threshold).

RESULTS: The framework was deployed across all seven health systems without any patient-level data leaving institutional firewalls. The cohort comprised 372,673 ICU encounters (mean age, 62 years; 45% female). In-hospital mortality ranged from 11% to 18% across sites (Table 1). Demographic composition varied substantially: the proportion of Black patients ranged from 2.4% to 59.1%, and the proportion of Hispanic patients from 2.1% to 18.8%. Among XGBoost models, cross-site validation yielded a median AUROC of 0.85 (range, 0.78–0.89); transfer learning, 0.86 (range, 0.79–0.89); and independent training, 0.87 (range, 0.85–0.89). Calibration improved with local adaptation: Brier scores ranged from 0.15 to 0.19 with cross-site validation and from 0.13 to 0.16 with independent training. The site with the poorest cross-site performance showed the largest gains. XGBoost AUROC increased from 0.78 to 0.85 with independent training, and ElasticNet AUROC increased from 0.77 to 0.82 with transfer learning alone.

CONCLUSIONS: A model-to-data benchmarking framework enabled novel prediction methods to be tested against diverse, private patient populations that would otherwise be inaccessible, providing more robust evidence of generalizability than public benchmarks alone. Across eight health systems, site-specific training consistently outperformed cross-site deployment, while transfer learning offered a practical middle ground by fine-tuning pre-trained weights on local data. By separating model development from evaluation infrastructure and data access, this approach lowers barriers to piloting innovative clinical AI methods, establishes clinically relevant benchmarks.

Table 1: In-Hospital Mortality Prediction: Summary Performance Across Seven External Sites

	AUROC		Brier Score	
Strategy	**XGBoost**	**ElasticNet**	**XGBoost**	**ElasticNet**
Cross-Site Validation	0.85 [0.78–0.89]	0.84 [0.77–0.87]	0.17 [0.15–0.19]	0.17 [0.14–0.19]
Transfer Learning	0.86 [0.79–0.89]	0.85 [0.82–0.88]	0.16 [0.14–0.18]	0.15 [0.14–0.17]
Independent Training	0.87 [0.85–0.89]	0.85 [0.82–0.88]	0.14 [0.13–0.16]	0.15 [0.14–0.17]
Lead Site (Internal)	0.88	0.87	0.13	0.13

Values represent median [range] across seven external sites. AUROC = area under the receiver operating characteristic curve. Brier score: lower is better (perfect calibration = 0). Lead site internal validation on held-out 2024 test data shown for reference.

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